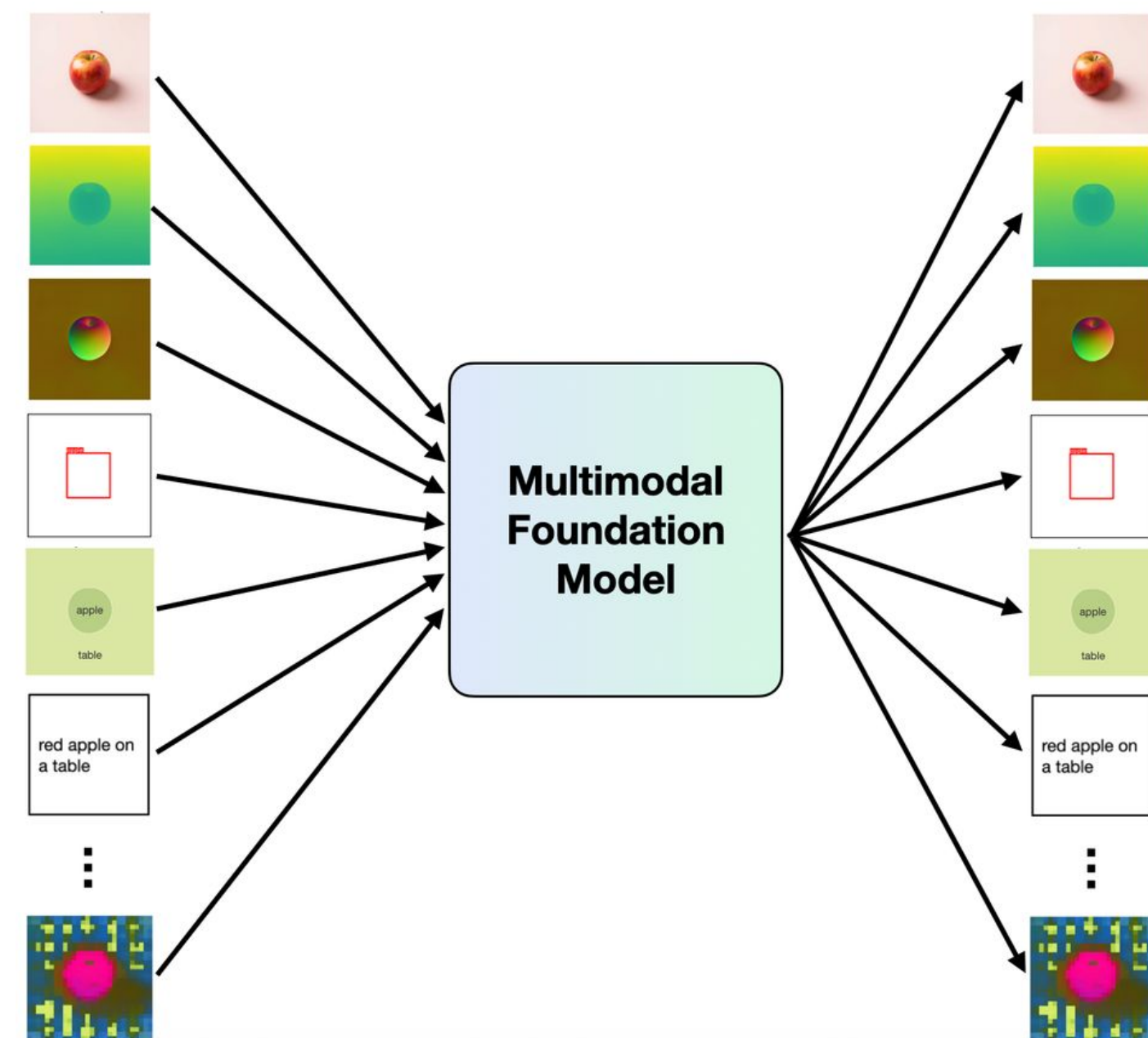


How Unified is your Encoder

Measuring Representational Convergence in Any-to-Any
Vision Models

Presented By :
Arthur Margeat, Adrien Clément,
Albert Fares, Martina Gatti



One trunk or many ?

- Each modality has **its own tokenizer** → converting raw values into tokens
- All modalities then pass through the same encoder, with **shared weights**

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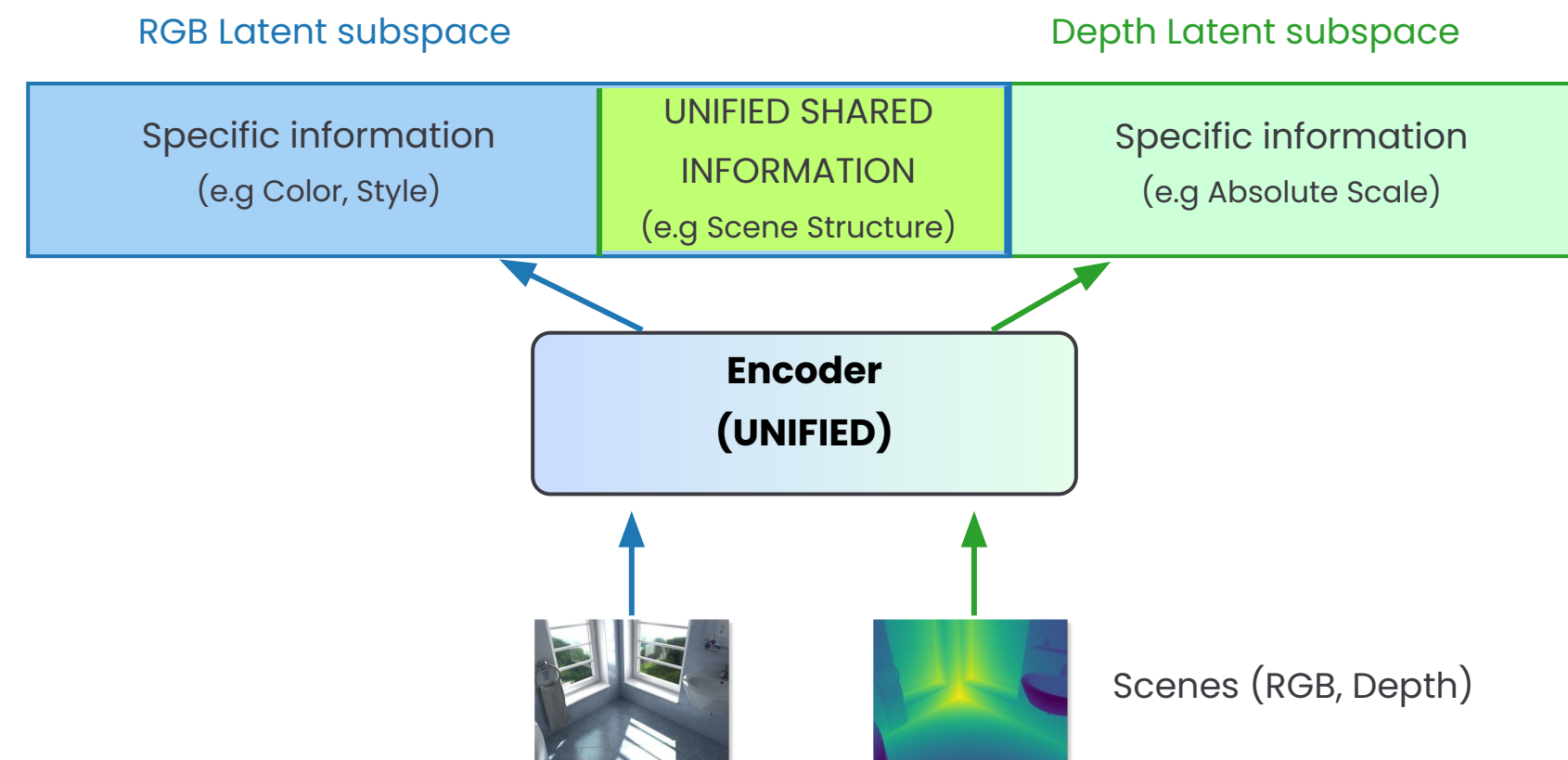
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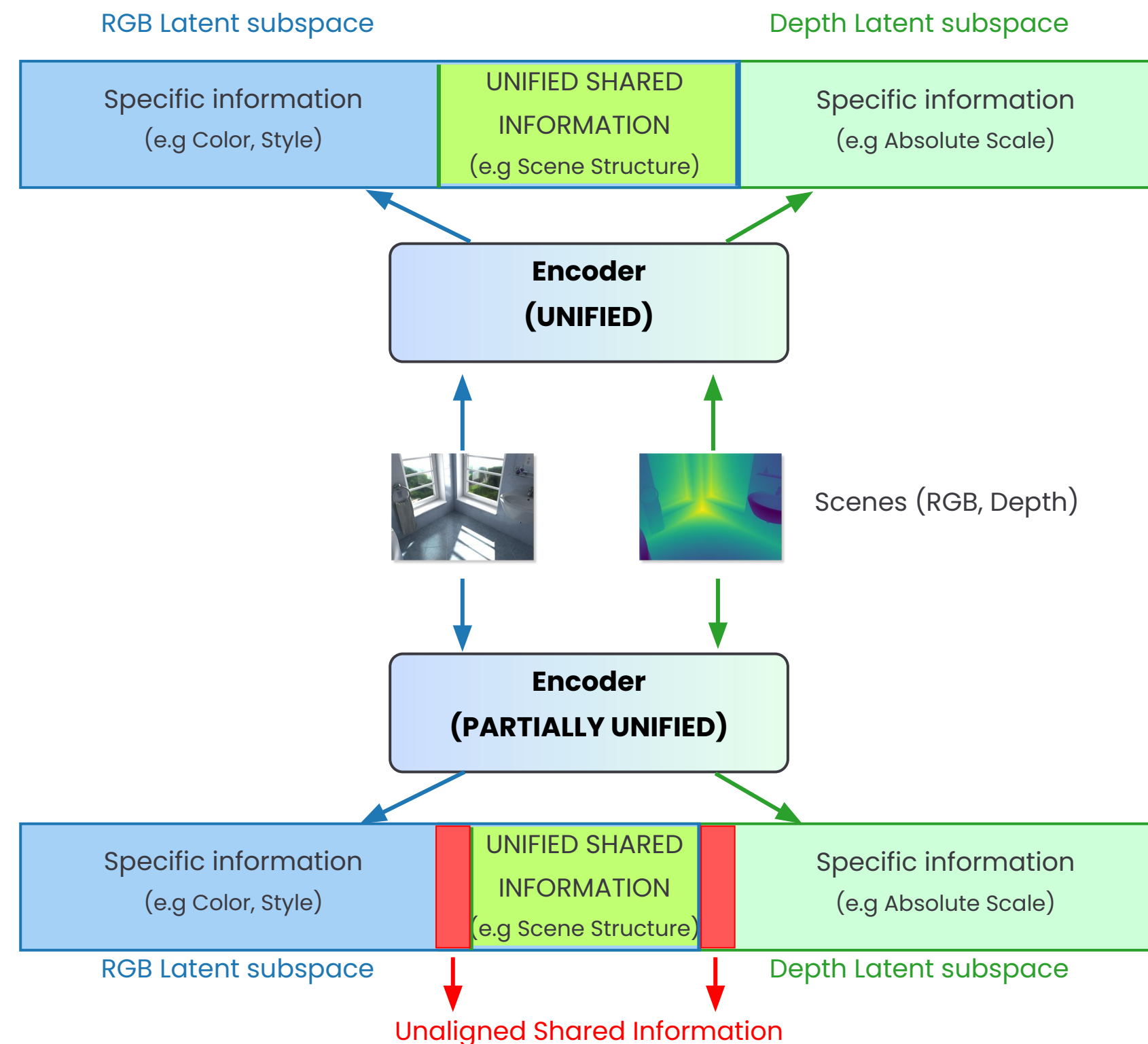


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Research question

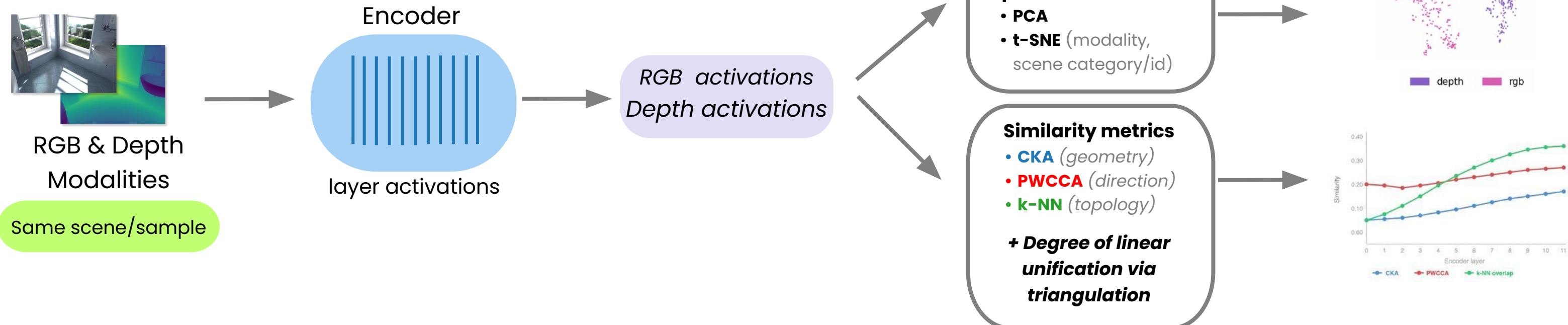
Does the 4M encoder build a single unified representation across modalities and if so, how complete is that unification?

“ We think that secretly, under the hood, the models cheat and create a little ensemble of independent models. [...] ”

Professor Amir Zamir

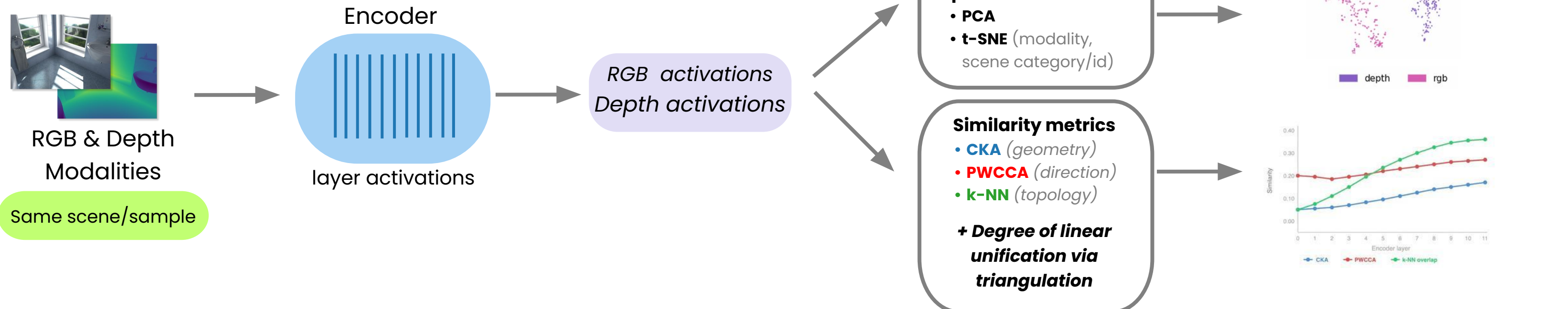
Methodology

Matched Modalities

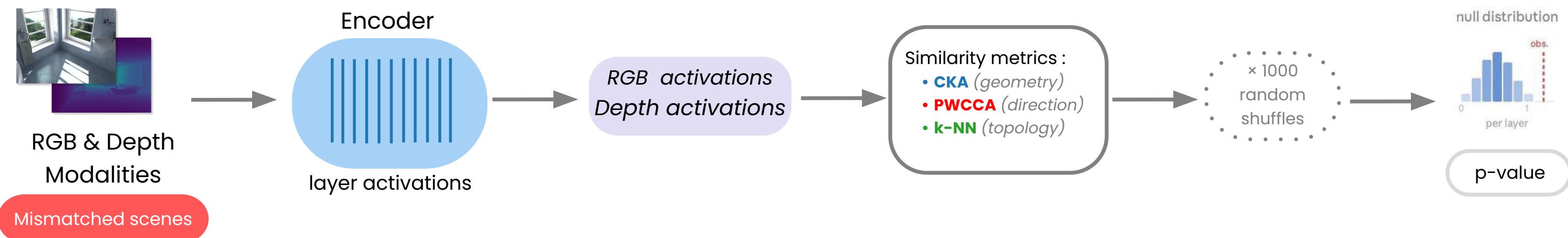


Methodology

Matched Modalities



Baseline : Null Hypothesis



Exploratory

Activation Space Geometry

analysis

the DNN activations visualized at each encoder layer using t-SNE

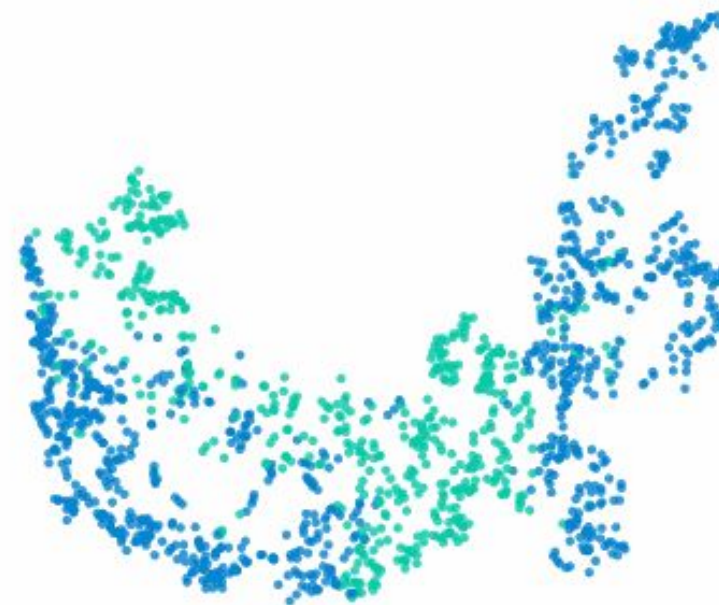
L00
encoder layer

Per modality



normals rgb

Per scene category



indoors outdoor

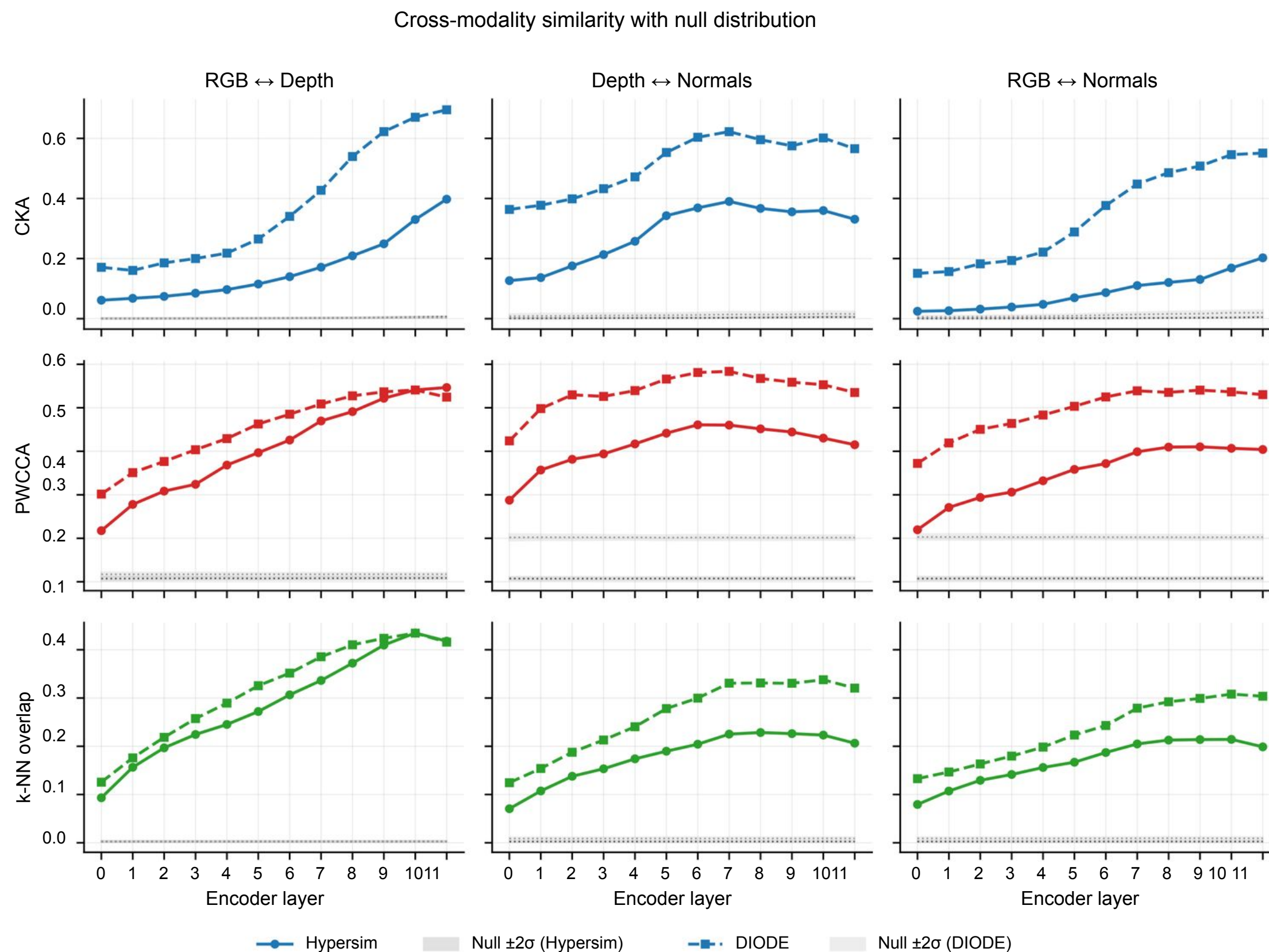
Per scene id



scene id (25 scenes)

Result

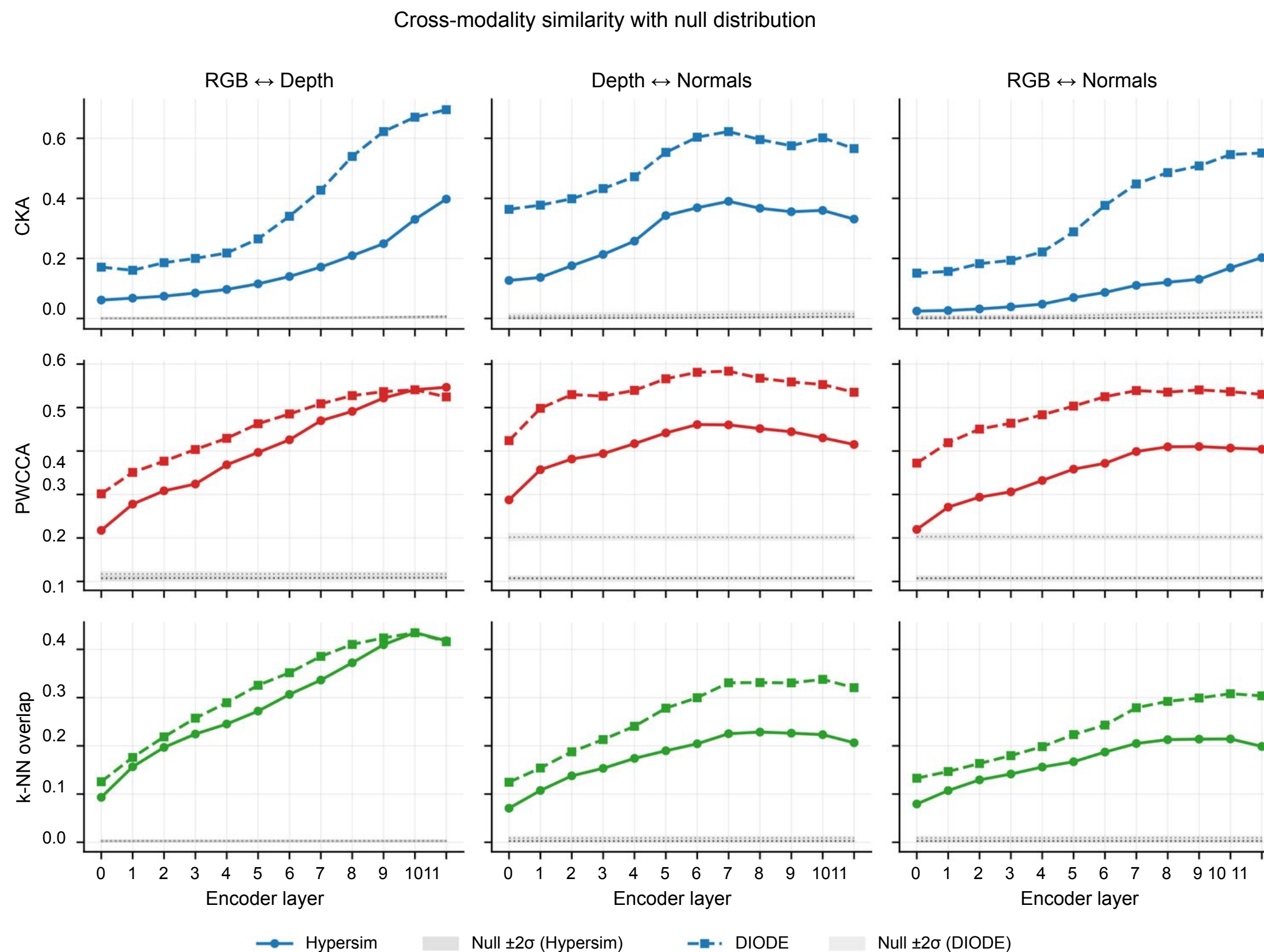
Based on Hypersim/DIODE Dataset, using
3 modalities, 4 000 samples total
p-value was minimal (0.004)



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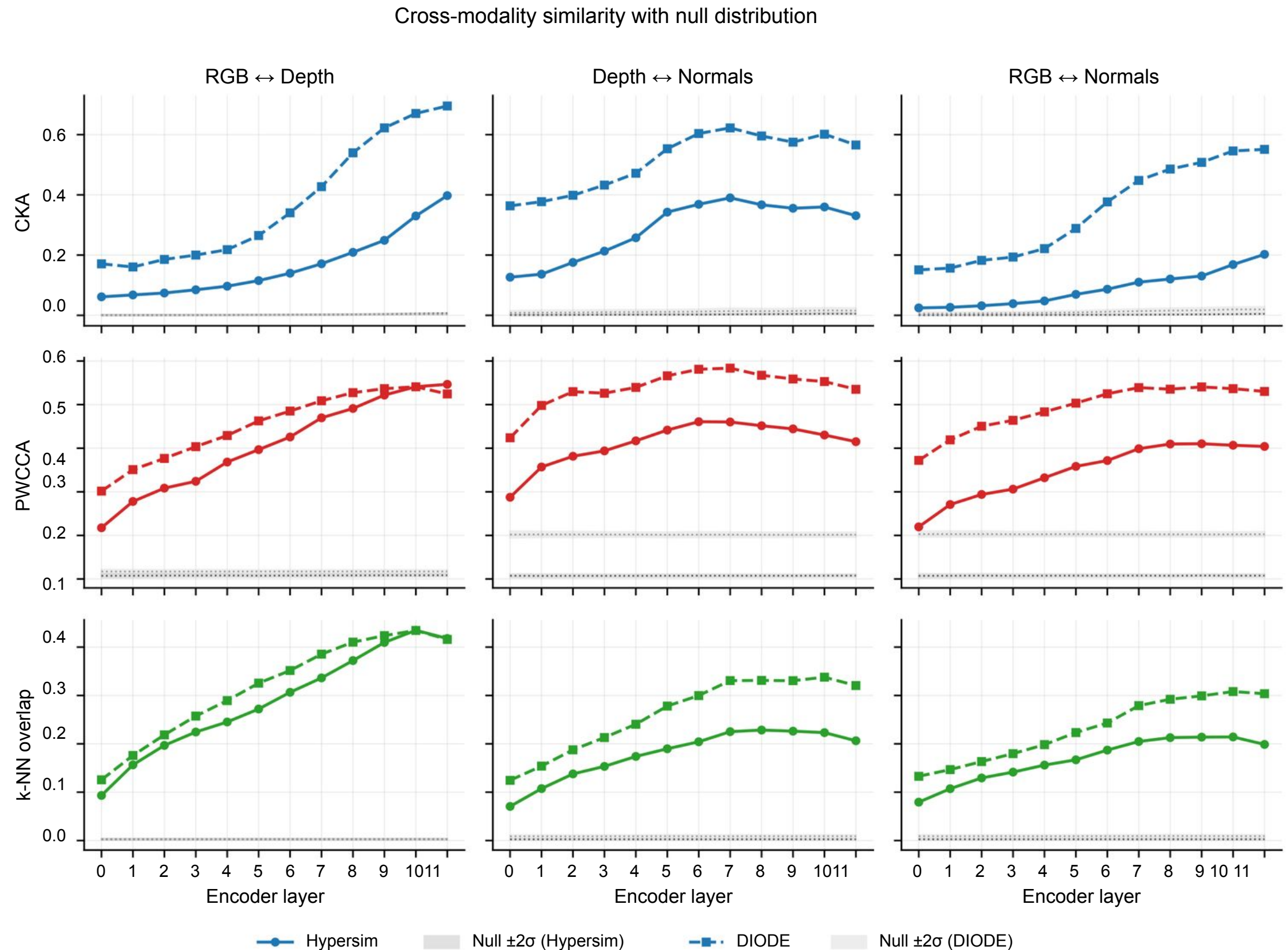
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1. Monotonic **rise across layers** for the 3 similarity metrics and modality pairs
2. Diode shows uniformly **higher similarity** than Hypersim (likely due to DIODE's lower visual diversity)



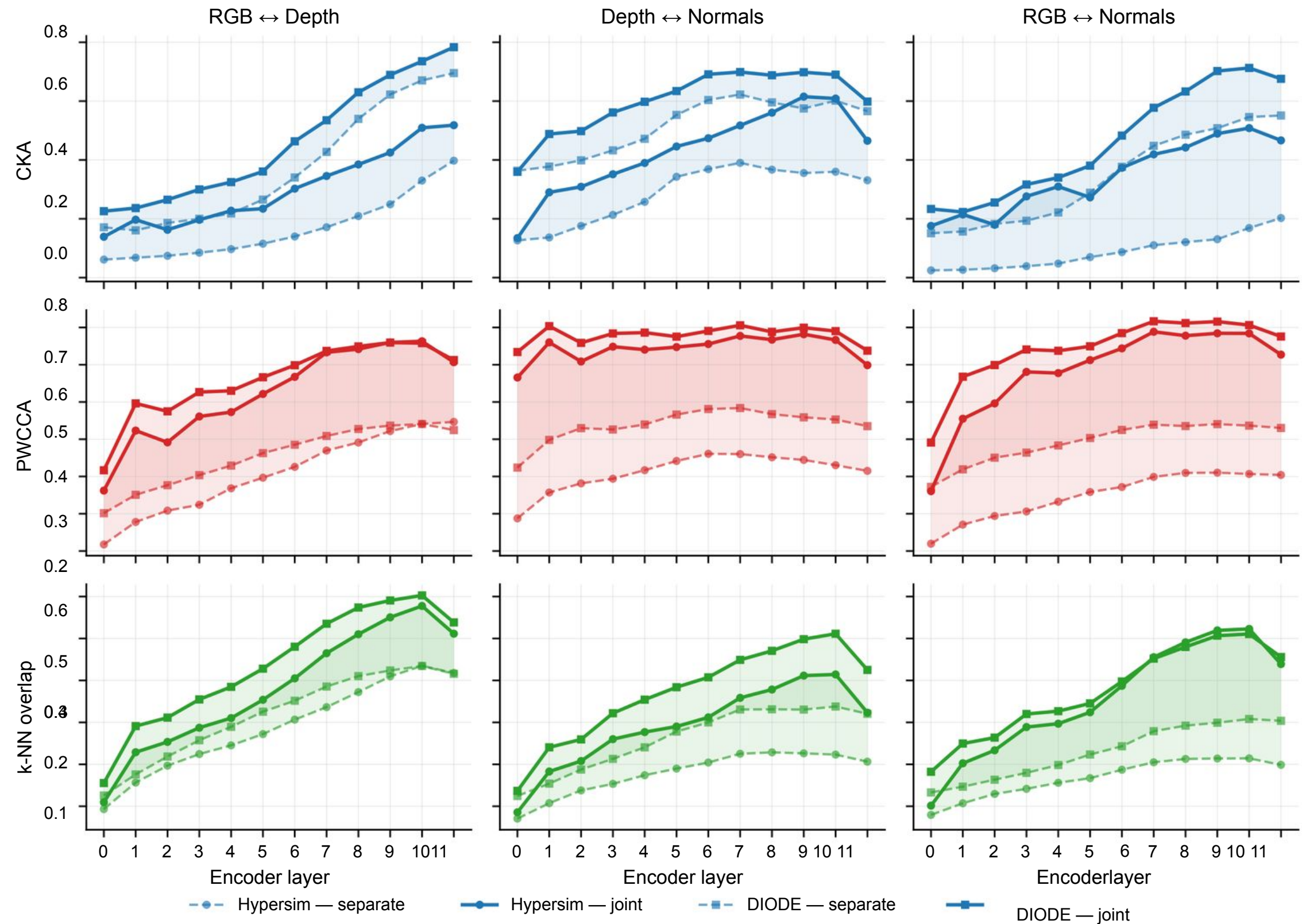
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Joint Pass : Both modalities concatenated into a single input sequence

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Joint pass (solid) vs separate passes (dashed) — shaded = gap joint processing closes



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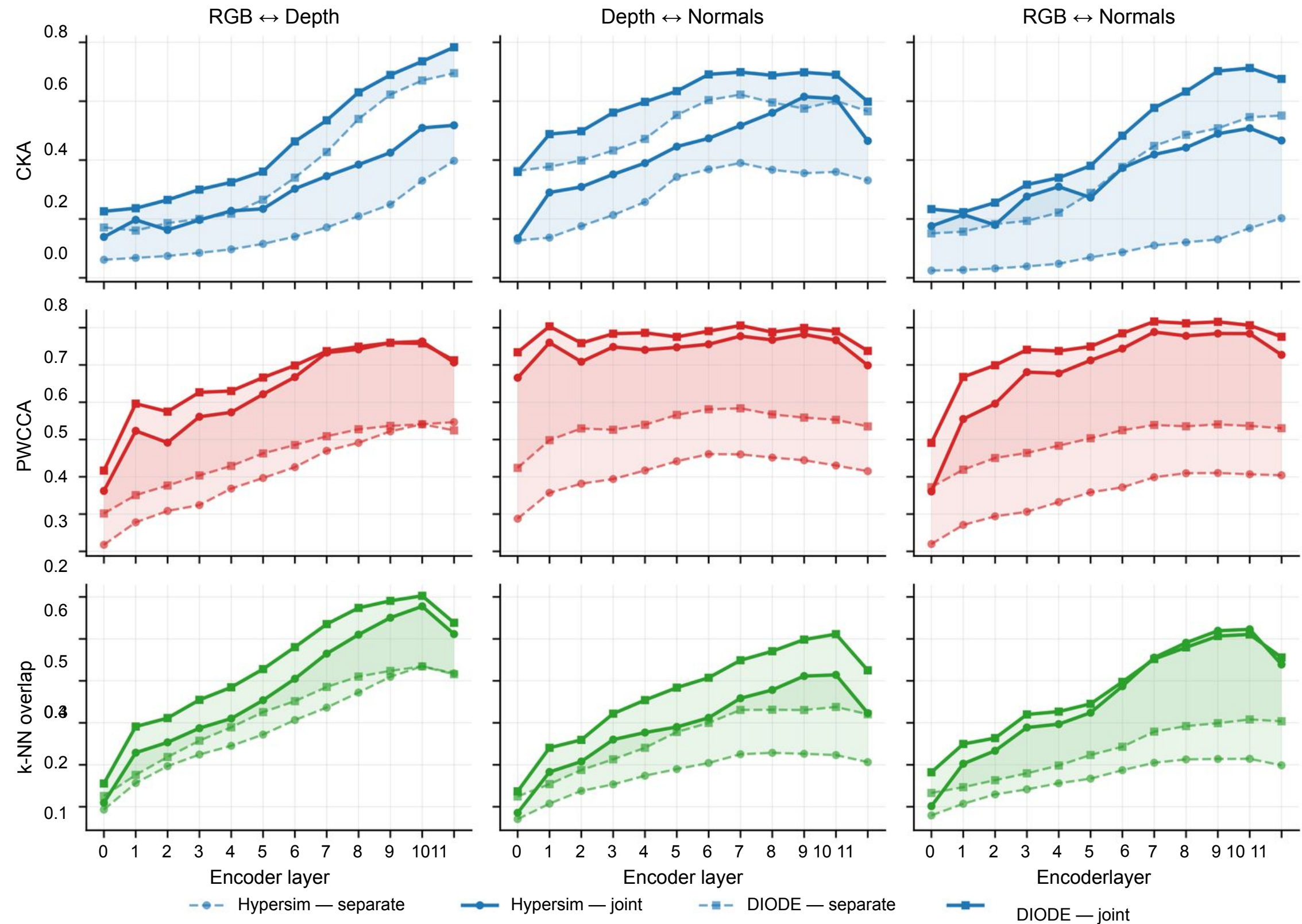
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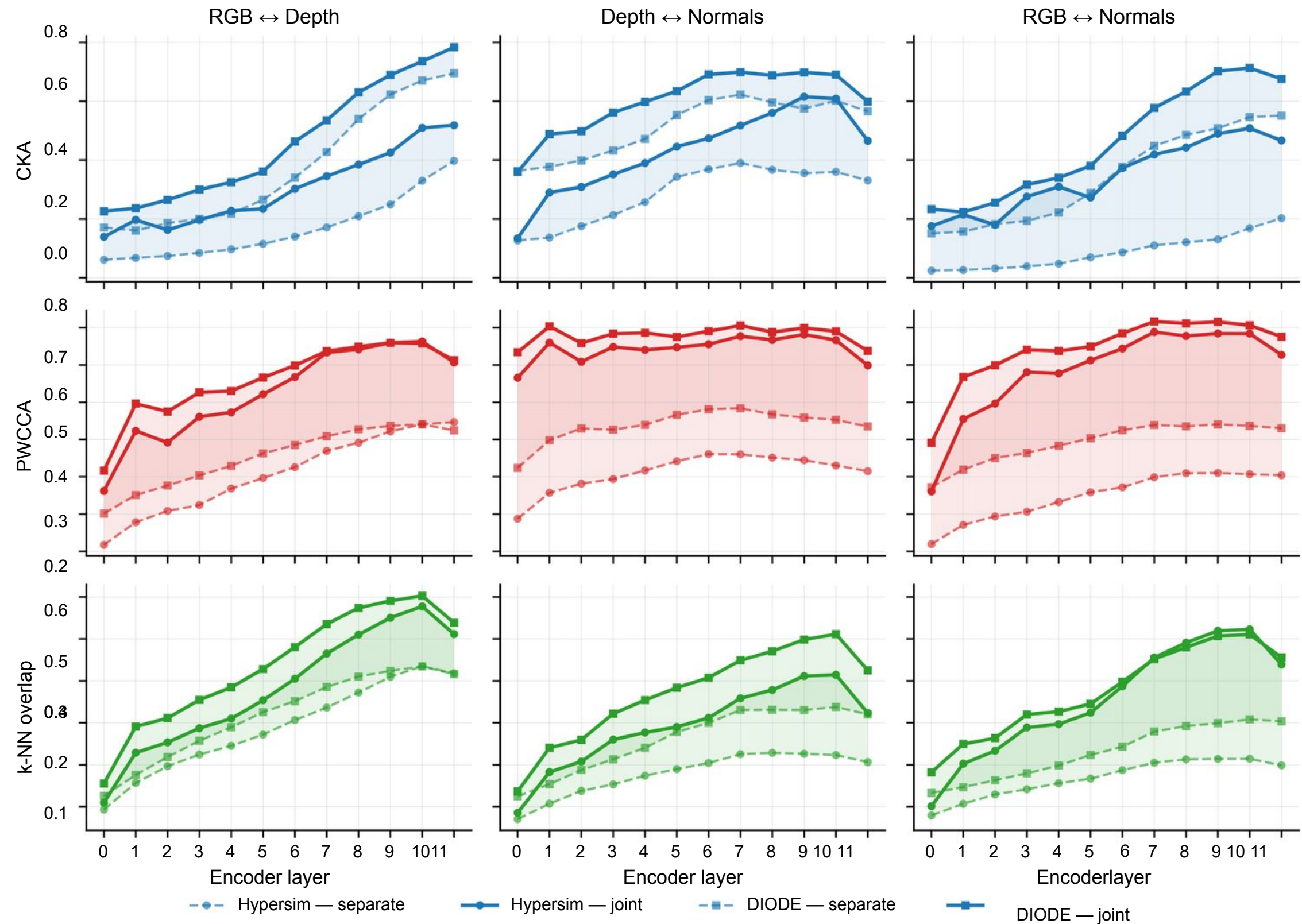
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2. The gain **concentrates at deep layers** and is largest for the hardest pair: RGB↔Normals

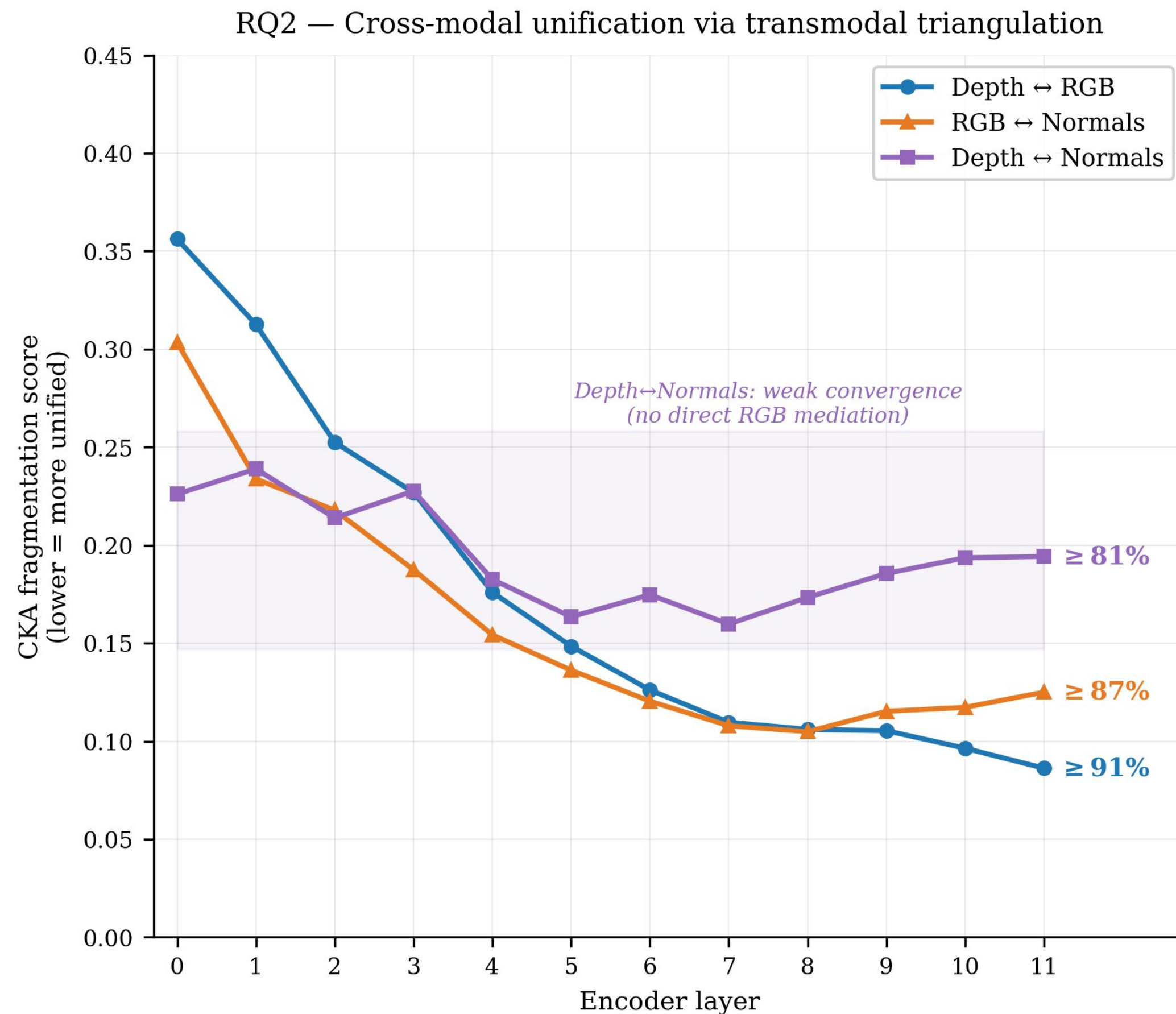
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Cross-modal unification (transmodal triangulation)

1. $\geq 86.5\%$ of cross-modal geometric structure linearly unified at layer 11 (avg. across pairs, under linear probing)



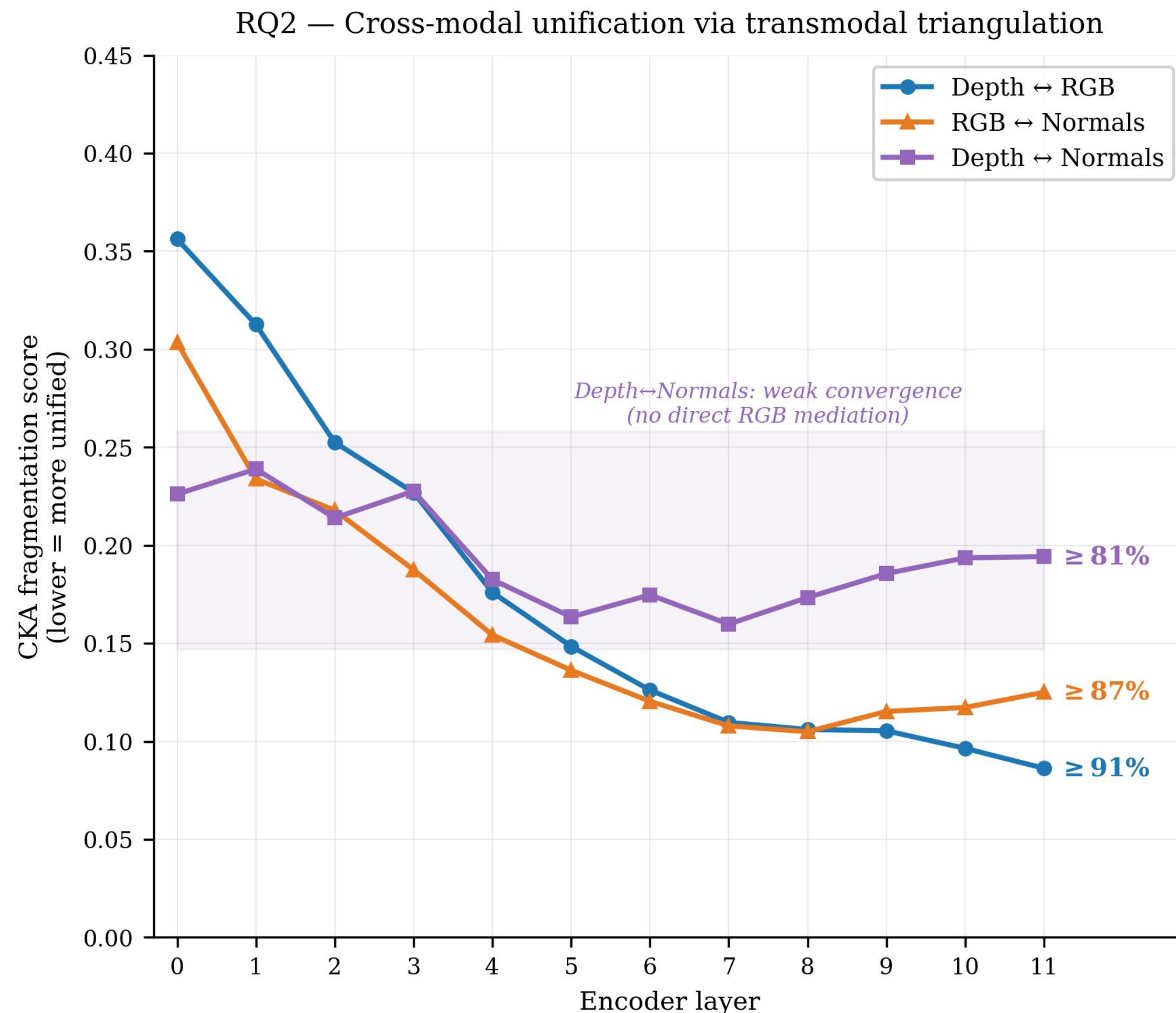
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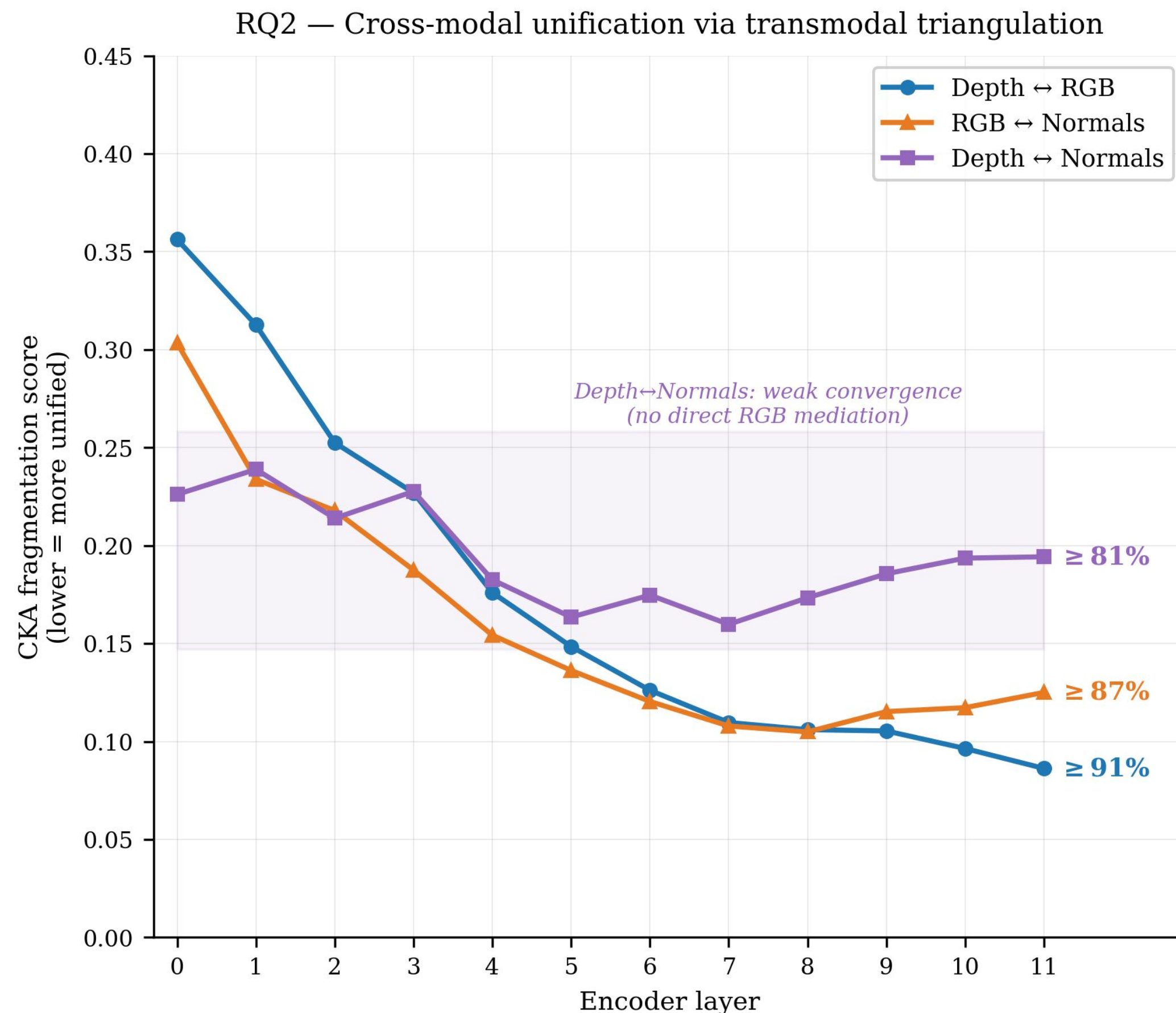
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3. Residual fragmentation > 0 across all pairs $\rightarrow$ **4M is measurably not perfectly unified** ($p < 0.001$)



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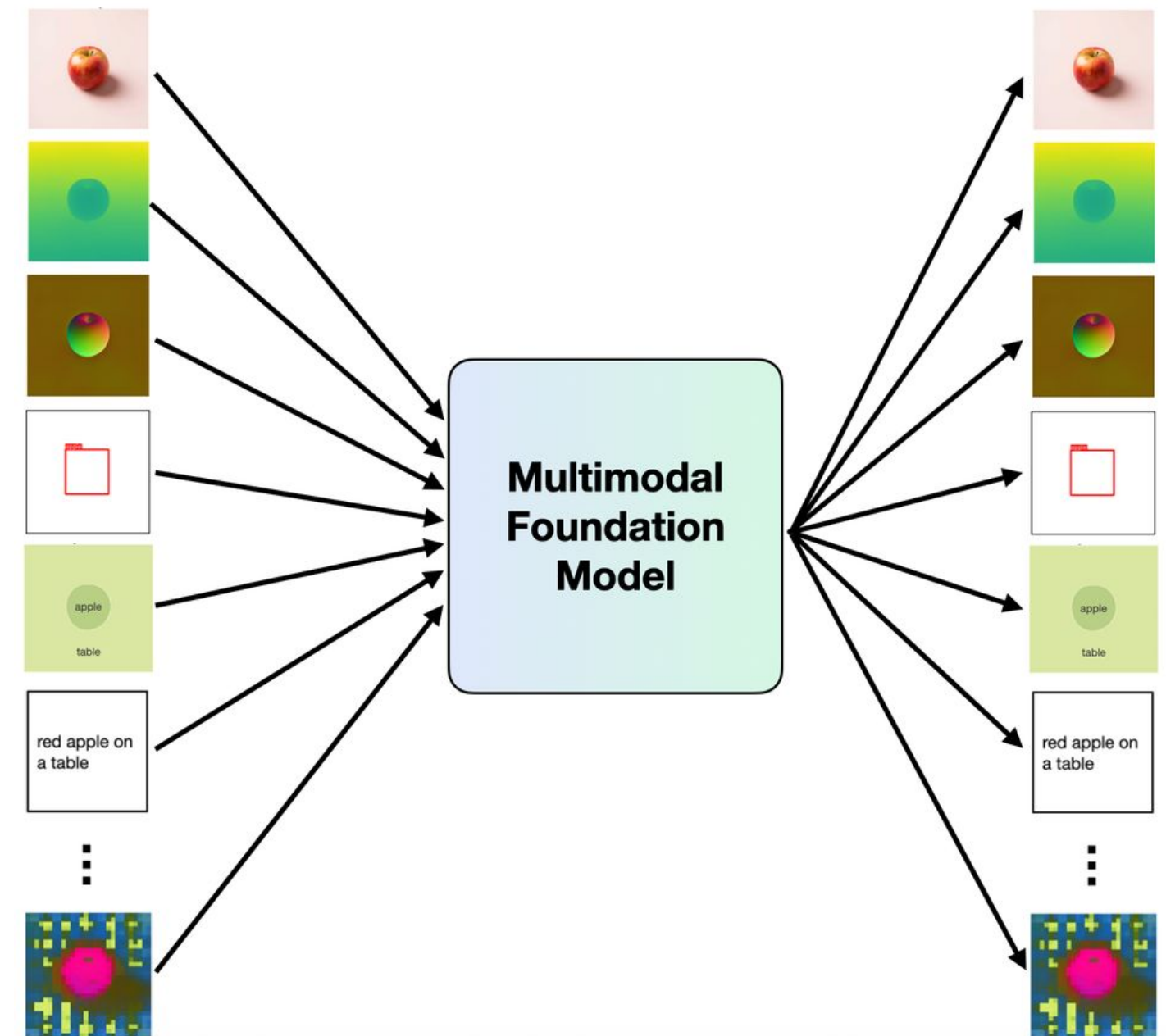
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3 Functional Compatibility test

- Cross-modal k-NN transfer and cross-modal decoding (functional compatibility)
- Training exposure confound (RGB seen far more often during training)

Thank You

Do you have any question ?



Results

Self-vs-joint conditioning: how much does the partner shift the representation?

